AI an Overview

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May, 2026

Philosophical Origins—Man as a Machine

- While the debate can be traced to the Greeks, it is in the seminal work of René Descartes (1596–1650) that we find the first systematic account of the mind/ body relationship.

- Philosophical questions introduced by Descartes are:

  - “What is the difference between a person and a machine?

  - Where is the line between the animate and the inanimate, between life and death?

  - Is there a difference between reason and ‘randomness’?”

Descartes’ “Treatise on Man” includes a comparison between a human being and a hypothetical “statue or machine” that operates like a clock or a hydraulic fountain.

Mechanical Computation

- The history of AI, which includes the philosophical considerations proposed by Descartes, would not be complete without consideration of mechanical computation.

Year	*Event*
30,000 B.C.E.	Paleolithic people in central Europe record numbers by notching tallies.
2600 B.C.E.	The Chinese introduce the abacus. (Used in China as recently as 1982.)
1500	Leonardo da Vinci invents the mechanical calculator.
1642	Blaise Pascal invents the first usable, hand-turned calculator for addition and subtraction (which he names *Pascaline*).
1670	Gottfried Leibniz enhances Pascaline with multiplication, division, and square root functions.
1822	Charles Babbage invents the Difference Engine (mechanical calculator), capable of adding and subtracting.
1936	Konrad Zuse creates a programmable digital computing machine (with vacuum tubes and a binary number–based architecture).

Events in the Development of Mechanical Computation

Historical and Philosophical Roots

- Alan Turing
  - Known for his work for the British military as a cryptanalyst (code cracker) during world war II
  - In 1946, he presented his seminal proposal for development in the mathematics division of an automatic computing engine
    - The essential aspect of computer presented such as:
      - Erasable memory,
      - Converters binary form of calculation to the decimal form out of input and output,
      - Logical control,
      - Central arithmetic part,
      - Means of selecting required information at a time.

Historical and Philosophical Roots

- The Turing Machine
    - Descendent of Babbage's Analytical Engine
    - Very simple in principle
    - An abstraction served to present the possibilities of AI
    - Had the ability to preform any computational process that can be performed
    - If a problem can be solved based on a set of instructions (algorithms) Turing machine can solve it
    - Hypothetical Turing machine is more powerful than a real machine because it has unlimited memory

Historical and Philosophical Roots

- The Turing Test: another Turing's contribution appeared in 1950
  - A paper titled "Computing Machinery and Intelligence" --- ideas on machine intelligence
  - The Criterion he proposed for deciding whether or not a machine is intelligent has come to be called the Turing Test.
  - The paper sparked a great deal of discussion and debate
  - The idea of the Turing test fits the behavior approach --- a machine is intelligent if it behaves like a human
  - Try to answer a question "Can machines think" using an imitation game

Historical and Philosophical Roots

- In Techniche festival in India, 2011, one version of the Turing test, Cleverbot, t involves judges having a short conversation with an open-domain chatbot or a human.. Both recorded in text.
- 30 volunteers a conversation with an unknown entity (50% with human and 50% with Cleverbot)
- When the other entity was a human only 63.3% believed it was a human
- When the other entity was Cleverbot, 59.3% thought it was human
- The secret of Cleverbot was, it stored information relating to hundreds of millions of conversation

Historical and Philosophical Roots

- Limitation
  - The Turing Test provides a limited assessment of AI’s intelligence and thinking ability
  - Conceptually -- the test fails to grapple properly with the complex issue of what is meant by “intelligence”
  - The Turing Test is limited because it produces a binary result

Man Vs Machine

- Some of the impressive achievements of AI in direct competition with human have been in the domain of complex games such as Chess, Go and Poker.

- What is especially impressive is that AI systems have convincingly the finest human exponents of these games

Man Vs Machine

- Chess
  - Garry Kasparov vs Deep Blue -- February 1996
    - Kasparov lost the opening game of the match
    - He won the next match
    - Third and Fourth ended in draw
    - Game five and six went to Kasparov
  - May 1997
    - The winner at this time was a machine

  - AI is now much superior than humans at chess.

Man Vs Machine

- Go
  - A board game of two players invented in China over 2,500 years ago.
  - A popular game with over 50 million players
  - It is phenomenally complicated and so poses a huge challenge for AI-powered machine.
  - Between 1985-2000, Taiwanese Ing Foundation offers a prize of $1,400,000 to any AI program that could beat a human champion.
  - Seems simpler than chess, but not really

Man Vs Machine

- Go
  - In 2016, Google DeepMind’s AlphaGo was the most advanced AI system for playing Go
  - AlphaGo analyzed 30 million Go moves from a game played by human players
    - Deep learning and reinforcement learning
  - AlphaGo opponent was Lee Sedol, an exceptional human Go player
  - The first three games was closely contested
  - AlphaGo became increasingly dominate the last three games and won

Man Vs Machine

- Shogi
  - Japanese game resembles closely resembling chess, but more complicated
  - In 2017 an AI system, Elmo, won human champion at shogi

- Pocker
  - Very complex imperfect information games
  - Winning poker involves using complex strategies so your opponent (human or AI) cannot decide whether you have a good hand or are bluffing. … no fixed strategy to win at poker.
  - Success depends on responding appropriately to complex interactions between players

Man Vs Machine

- Pocker
  - A much-publicized poker competition between human and AI took place in January 2017
  - Human race represented by four top poker professionals while AI represented by Libratus.
  - Initially Libratus given the rules of poker, then played trillions of games and progressively improved its playing
    - Use reinforcement learning
  - All four human players lost to Libratus

Man Vs Machine

- Pocker
  - Libratus won one player in a game against only one other player at a time
  - More impressive if an AI system could beat several top human players at once.
  - Accordingly, an improved version of Libratus called Pluribus was produced
  - Pluribus has faster algorithms and is generally more efficient

Human Limitation

- Human has many cognitive strengths but also many cognitive limitations
  - Limited processing capacity--- attend a limited number of things at any give time
  - Possess numerous cognitive biases – tendency to ignore information
  - Ability to use our cognitive abilities effectively is severely compromised by various emotional states
    - Emotional states of anxiety and depression impair our ability to think

Human Limitation

- Limited capacity
  - Human information processing is slower than that of AI-powered machines
  - The world most powerful computer, in 2018, performed 200,000 (200 petaFLOPS) trillion calculations per second
  - While human can keep four items in mind at time
  - Human information processing speed is remarkably low …. 0.01 Flops per second

Human Limitation

- Sustained attention

    - Vigilance decrement often occurs because people become less alert when performing repetitious tasks

    - Poor sustained attention is also dangerous in car driving… drivers engage in mind wandering 63% of the time and actively focused on driving only 15-20% of the time

    - Sustained attention in Formula 1 drivers

Human Limitation

- Forgetting
  - Failures to retrospective (memory for past events and information) and prospective (remembering to perform some intended action) memory
- Mental set
  - Fails to solve problems efficiently because over-influenced by past experience

How intelligent is AI

- Moravec's Paradox
  - Moravec proposed his famous paradox---"It is comparatively easy to make computers exhibit adult level performance on intelligent games and playing pokers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility"
  - Why you asked to do CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart)---- to insure a website user is a human… automated computer system may not solve such challenge
  - Nechar et al. (2015) devised a sophisticated AI program focusing on edge corners detection--- the performance was 57%
  - Visual perception is complex ---- most complex visual process are largely unconscious because they have developed gradually over time

How intelligent is AI

- Moravec's paradox is also applicable to human motor skills and abilities --- unconscious too

- Zador (2019) pointed out that "We cannot build a machine capable of building a nest, or stalking prey…. In many ways, AI is far from achieving the intelligence of a dog or a mouse, even of a spider"

- These species high structured brain connectivity that facilitate the learning of complex forms of behavior --- encoded in the genome of these species

How intelligent is AI

- Intelligence: Refers to a very general ability to solve an enormous number of very heterogeneous problems
    - Main criteria for human possess high intelligence
        - The acquisition of a wide range of new knowledge and skills
        - The mastery of language including the ability to engage in meaningful conversations
        - The position of good short-term memory
        - Using pre-existing knowledge and skills to facilitate learning on novel tasks
        - The acquisition of abstract knowledge by generalizing from more concreter form of knowledge
        - The management of several conflicting goals and priorities
        - The possession of emotional intelligence

How intelligent is AI

- Human Intelligence
  - Intelligence test (IQ test) measure general intelligence because they assess a wide range of abilities including mathematical ability, verbal ability and reasoning
    - Anything 140 or above considered genius territory
  - These test do not included the practical and social skills required for success in life

How intelligent is AI

- AI and Intelligence
  - Majority of AI systems are narrow intelligence rather than artificial general intelligence
  - Deep Blue and AlphaGo would not perform well on other cognitively demanding tasks
  - Intelligence test
    - An intelligent system should possess four characteristics
      - The ability to acquire knowledge
      - The ability to master knowledge
      - The Ability to create knowledge
      - The ability to output knowledge to the outside world

How intelligent is AI

- Intelligence test (IQ)
  - The average human adult has an IQ 100. …. Six-year old has an IQ 55

Rank (Approx. Gen)	Model / AI	Estimated IQ Equivalent (Standard Human Scale)
Top Tier	Google Gemini 3 / OpenAI GPT-5 / Claude Opus 4	**140 - 150+** (Near-genius level on verbal/reasoning tests)
Mid-to-High	Open-source models (Llama 4, DeepSeek V4)	**120 - 130+** (Superior human capability)
Voice Assistants	Modern Apple Intelligence / Siri / Google Assistant	Tied directly to the cloud models above, elevating them past the 100+ "average human" mark.

How intelligent is AI

- Language
  - There are close links among intelligence, thinking and language ability
  - Humans with excellent language skills have greater general intelligence and thinking power than those with inferior language skills
  - AI is successful in various language skills
    - Speech recognition
    - Answering general knowledge questions
    - Translating
    - Holding conversation
  - However, human-like language performance is limited

How intelligent is AI

Skill	Former State (Pre-2023)	Current State (2025–2026)
Speech Recognition	Struggles with noise, accents, and medical terms.	Sub-3% Word Error Rates; highly capable.
General Knowledge	Good at basic trivia; struggled with deep reasoning.	Matches or exceeds human experts on USMLE exams and GPQA.
Translating	Clunky, literal, often lacked cultural context.	Often preferred over professional human translators for high-resource languages.
Conversation	Slow, mechanical, easily confused by topic shifts.	Native audio-to-audio processing under 300ms latency; utilizes long-term memory.

How intelligent is AI

- Adaptation to environments
  - Intelligence typically is defined as consisting of adaptation to the environment
  - In 2019, a competition was organized called Animal-AI Olympics …. opened to 60 AI systems with prize money of $32,000 for those systems exhibiting the best performance

  - Performance was assessed on 300 tasks in 12 different categories

  - Only AI system having reasonable perceptual, motor, and reasoning ability could perform the various tasks successfully.

  - The winner was Trrrrr with 43.6% overall performance, Sirius with 38.7%
  - All worst 22 AI systems averaged under 10%
  - Even the threes best performers struggles in some tasks

How intelligent is AI

- Creativity
  - More creative form of thinking where there are numerous possible solutions to relatively open-ended problems
  - Various types of creativity
    - Combinatorial creativity – finding solutions for apparently unrelated ideas
    - Exploratory creativity – more complex than combinatorial creativity– involves variations in themes… e.g Jazz Musicians, artists
    - Transformational creativity – involves creative ideas more profound, novel, and surprising than those associated with exploratory creativity --- very rare in human. E.g Einstein's relativity theory ---- no research is tested its presence in AI system
  - AI has shown combinatorial creativity
  - AI have also exhibited exploratory creativity e.g AI system to learn Jazz music

How intelligent is AI

- Art
  - October 25, 2018 was landmark in AI history
    - AI-created painting called “Portrait of Edmond de Belamy” was sold at Christie’s in New York for over 300,000 pound

- Initially the AI system provided with 15,000 portraits.
- A generative adversary network was used
- The generator created new AI portrait based on the dataset then the discriminator tried to detect difference between that portrait and a human-created portrait
- The process repeated so many times until the discriminator decided a given AI portrait is painted by a human being

How intelligent is AI

- Mazzone and Elgammal (2019) made use of more sophisticated techniques
  - Initially provided by 80,000 image of painting from the five centuries of western art history
  - The discriminator instructed the Generator to follow the aesthetics of the art it had been shown but not emulate any already established style
  - These conflicting instruction caused a dynamic conflicting produce the art that was somewhat novel

How intelligent is AI

- Mazzone and Elgammal (2019)….
  - Visual Turing Test
    - Tested the artistic merit of their AI-created arts by seeing whether human viewers test the difference between it and human created art.
      - They found 75% of AICAN-created art was thought to be human-created

- A common criticism of such finding is that the AI-generated paintings are only minimally creative because they depend so heavily on human-generated paintings.

How intelligent is AI

- How much credit for AI-created art should go to the AI system … three main reasons for skepticism

    - Un-like AI artists are driven by various goals

    - Human artists play an important role in the creation of AI art

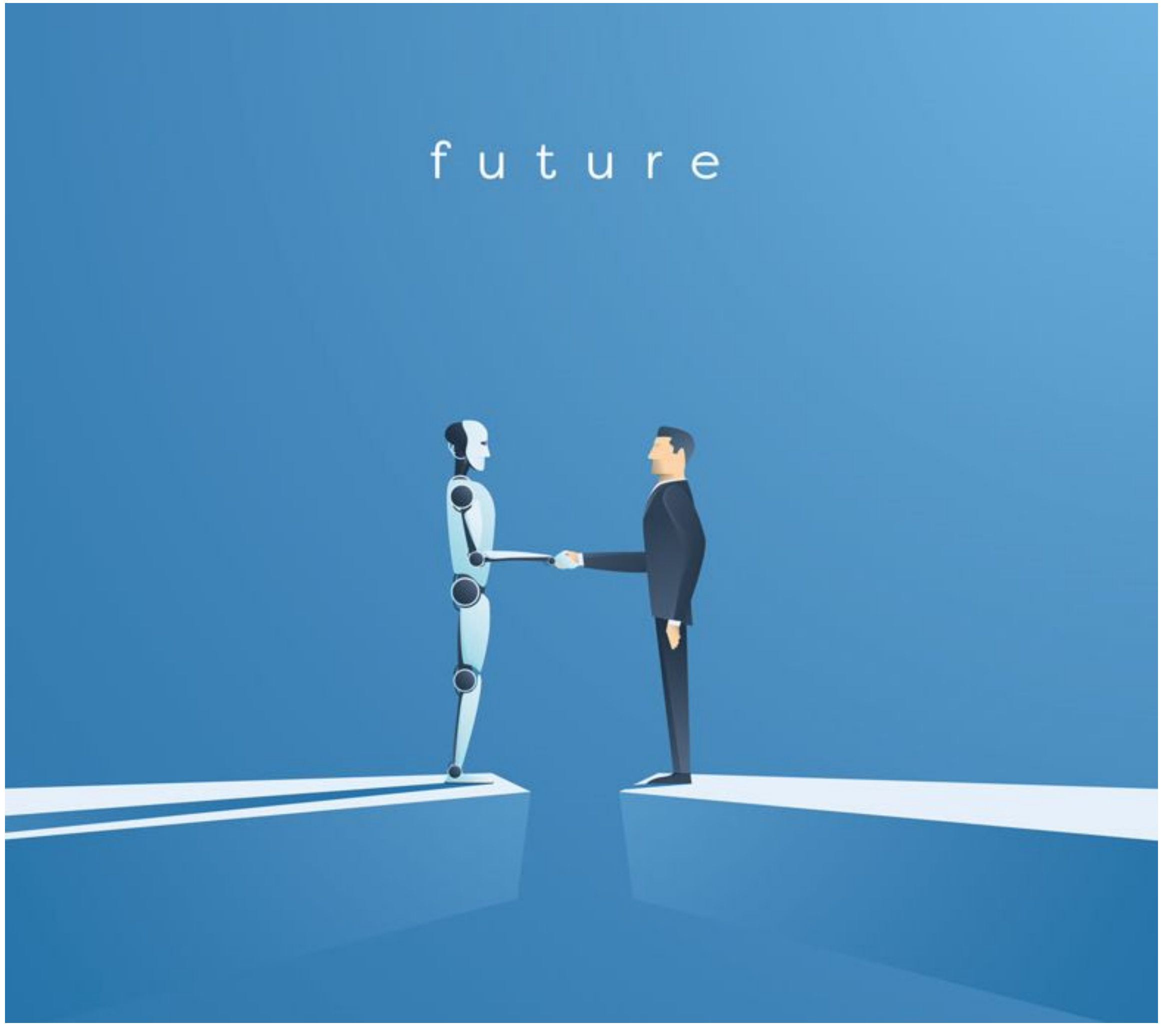
    - There are several stages in creative process

Marriage of AI and Humans

- Human intelligence is:
  - Flexible
  - Commonsensical
  - Imprecise
  - Emphatic
  - Creative
- AI processing
  - Consistent
  - Fast
  - Efficient
- Smith 2019, AI excels at calculation whereas humans excel at judgement

"While some may perceive AI to be a threat to humans, we believe the true potential of technology lies in humans and machines working together."

Marriage of AI and Humans



Marriage of AI and Humans

- The differences between human and AI suggest there could be huge advantages in combining and integrating their respective strengths.
  - Create hybrid intelligence… combines human and AI intelligence to produce performance superior to either
  - Possible through bi-directional learning
    - Establish a two-way direct communication link between the brain and the external world

Brian-Computer Interface

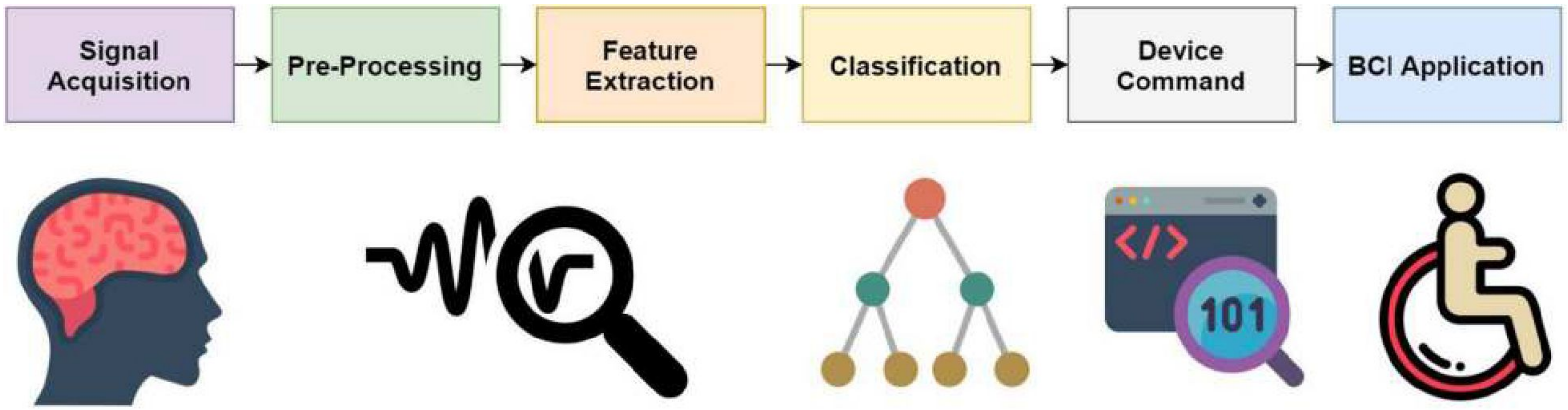
- Brain-Computer Interface (BCI) aims at controlling different assistive devices through the utilization of brain waves….. called brain-machine interface
  - Efficiently use the brain’s electric signals to maneuver external devices

- An effective as well as a powerful tool for user-system communication.

- *“A hardware and software communications strategy that empowers humans to interact with their surroundings with no inclusion of peripheral nerves or muscles by utilizing control signals produced from electroencephalographic activity.”* (Rashed et al., 2020)

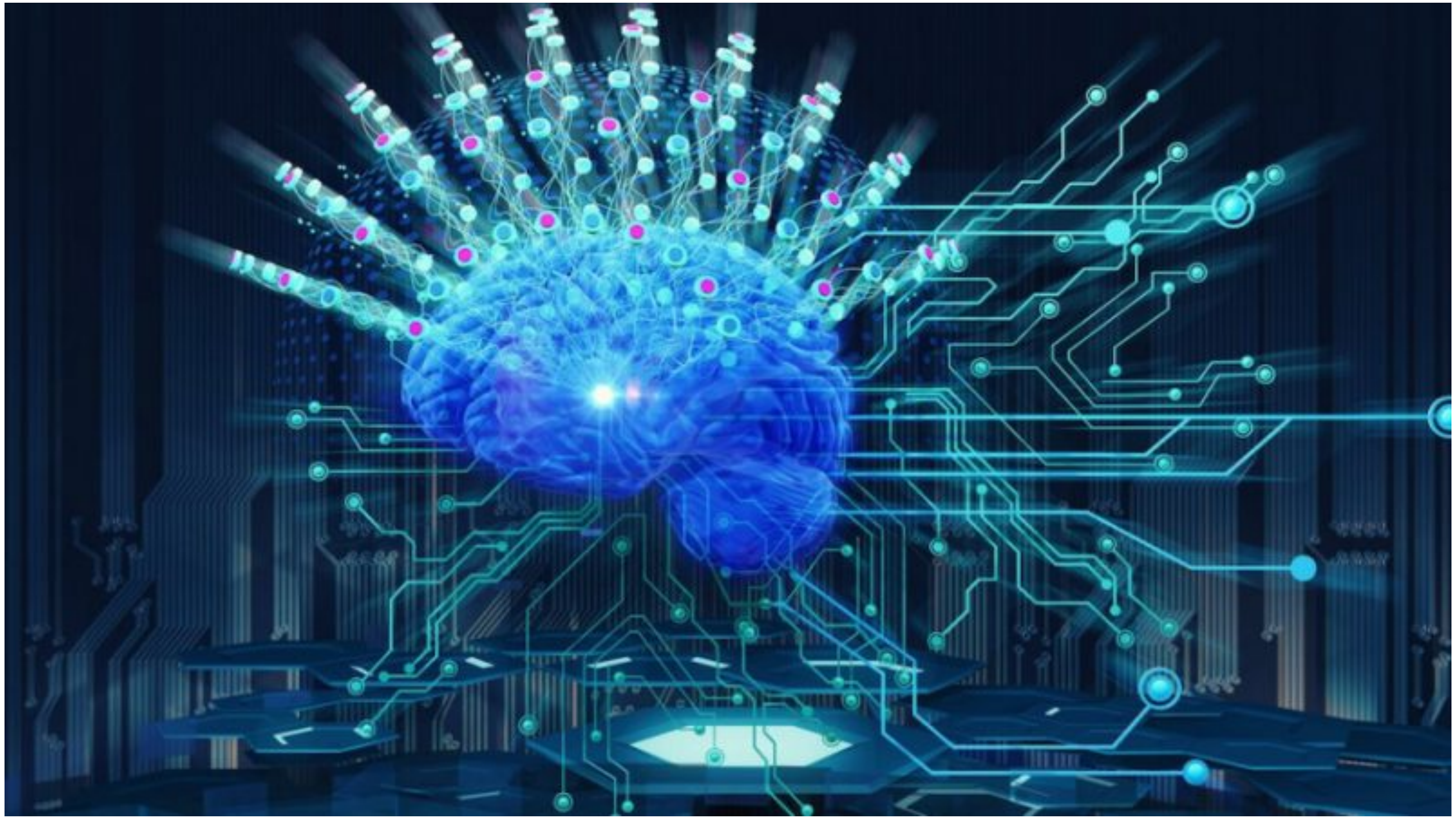
Brian-Computer Interface

- Essentially components:
  - Brain activity measurement: captured through the various types of EEG sensors
  - Preprocessing: eliminate artifacts
  - Feature extraction: selection of significant features
  - Classification: features classified using machine learning algorithms
  - Translation into a command: classified outcomes translated into device commands to develop real-life BCI applications



Brian-Computer Interface

What is the belief behind CBI? Why humans work to realize the CBI?



Brian-Computer Interface

"The belief behind these endeavors is based on the perception that humans are confronted by an "existential threat" in the form of artificial intelligence which can be mitigated by merging with the aforementioned threat." [Elon Musk]

"While some may perceive it to be a threat to humans, we believe the true potential of technology lies in humans and machines working together." [Intelematics]

"BCIs, which represent technologies designed to communicate with the central nervous system, and neural sensory organs can provide a muscle independent communication channel for people with neurodegenerative diseases, such as amyotrophic lateral sclerosis, or acquired brain injuries"

Brian-Computer Interface

- Neuralink, Elon Musk’s company has constructed a self-contained neural implant which can be used to transmit comprehensive brain activity remotely.

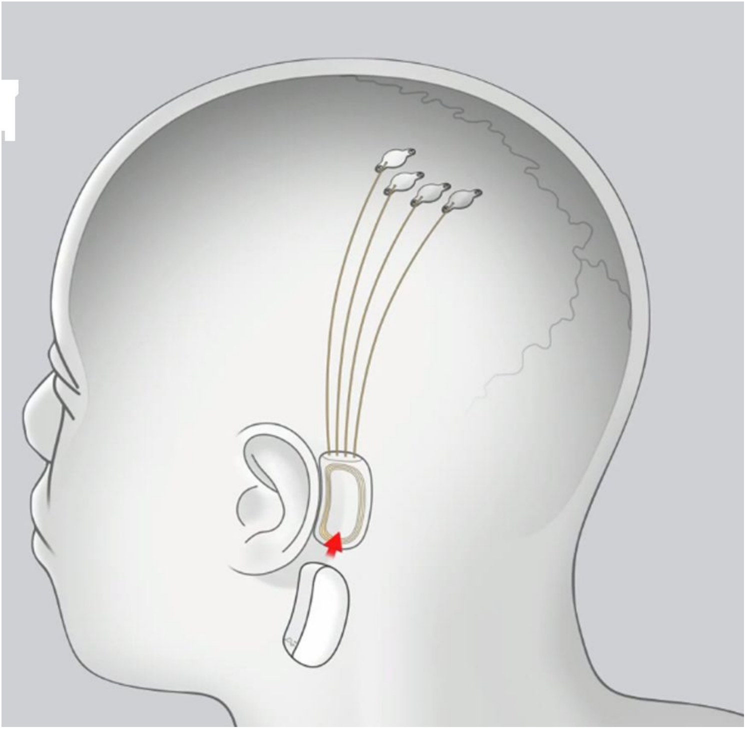
- The implant can be controlled via a computer or mobile devices referred to as the tertiary layer of cognition through the neuralink connected by Bluetooth low energy antenna to an app.

- The battery of the implant is to be charged wirelessly through a compact inductive charger.

- Micro-scale threads are implanted into regions of the mind that command movement.

Brian-Computer Interface

- Threads are connected to a chip which is further linked to a small computer behind the ear called the link which houses software as well as the battery.
- Neurosurgery requirements of the procedure to insert the neuralink without general anesthesia for the activity of the brain to be observed by an exterior receiver.
- Thickness of the neuralink device is 8mm



Brian-Computer Interface

- Neuralink’s potential end goals are curing:
  - Hearing impairment,
  - Neurological disorders
  - Alzheimer’s,
  - Dementia,
  - Blindness,
  - Paralysis,
  - Extreme pains,
  - Seizures
  - Addiction,
  - Detect Strokes and brain damages

Robots: As Example

- Science fiction has encouraged us to think of robots as systems mimicking, with significant success, the behavior of humans.

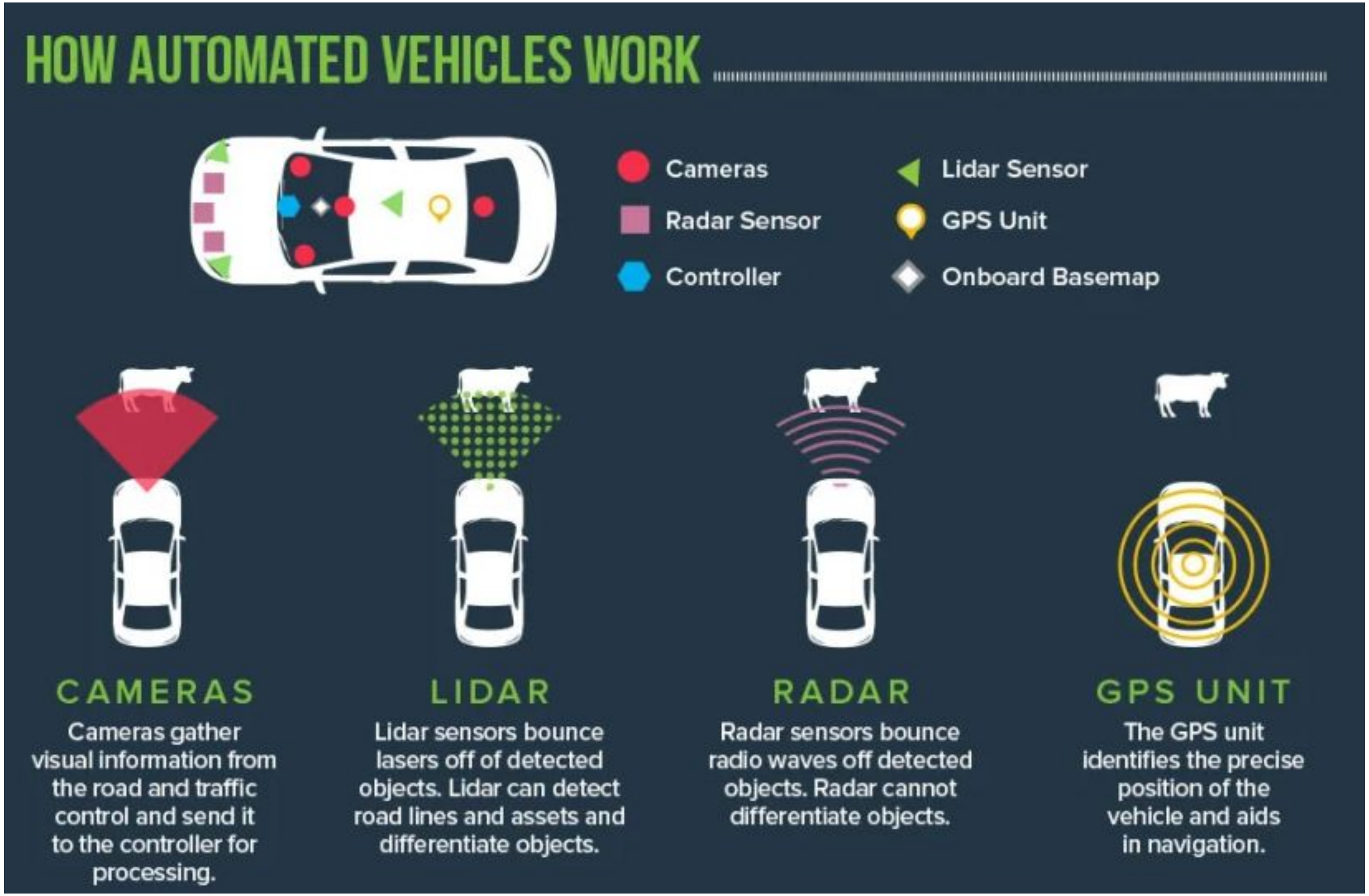
- Robots perform specific functions
  - A robot to carry out stereotyped assembly-line chores or
  - A vehicle that can get where it needs to without human driver

Robots: As Example

- There are over 3 million robots worldwide
- Autonomous vehicles
  - In 1939, General Motors presented the first-ever autonomous car at the New York World's Fair. Was
    - Powered by electricity
    - Steered by radio-controlled electromagnetic fields
  - Why?
    - Currently there are about 1.3 million traffic deaths worldwide in one year
    - Traffic accident happened mostly due to human error
    - "Autonomous vehicles can easily avoid all these"

There are questions on how autonomous vehicles should behave, and how responsibility and risk should be distributed in the complicated system the vehicles operates in.

Robots: As Example



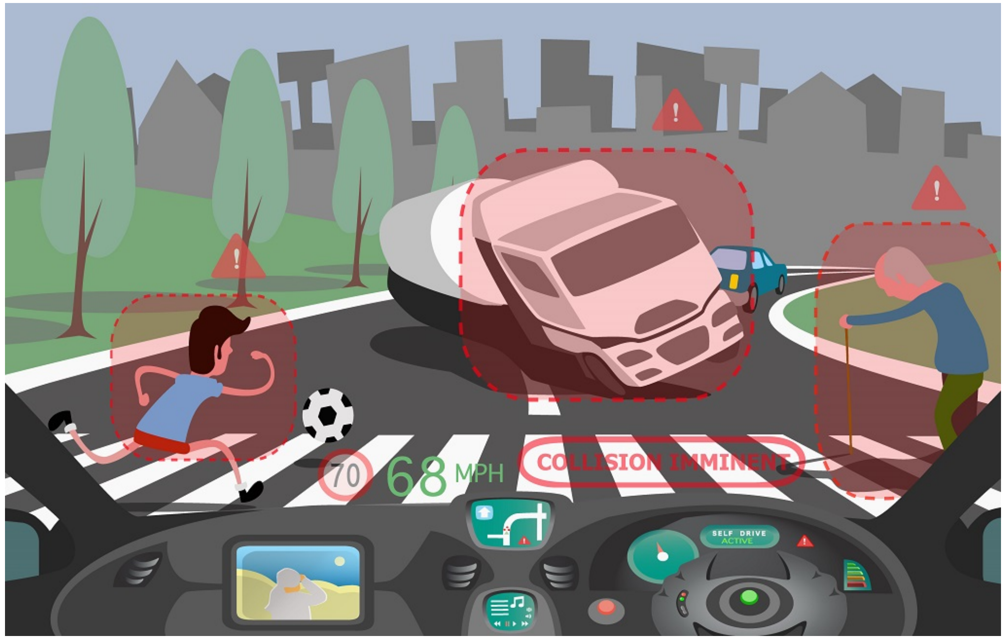
Robots: As Example

- Koopman et. al (2019) mentioned that autonomous vehicles need to be programmed to respond appropriately to novel situations. …. many autonomous vehicles fall short of fulfilling the criterion.

- “In the future autonomous cars will become safer than human-driven ones”

Robot and Morality

- Ethical issues
  - There is a debate among politicians, AI experts, psychologists, and philosophers on autonomous vehicles…. Should be programmed ethically.
  - The “trolley problems” used as a theoretical tool to investigate ethical intuitions and theories
  - Jeremy Bentham (1748-1832)…. Utilitarianism…. The greatest good for the greatest number



Robot and Morality

- Warfare-autonomous weapons …. killer robots and other lethal AI systems
  - Robots and other autonomous weapons systems will play a central role in the future wars.
    - In 2008, there were between 4,000 – 6,000 ground robots in Iraq…. Used to detonate roadside explosive devices
    - Since then MQ-9 Reaper unmanned aerial vehicle has been developed …. As a killer robots
  - Role of AI in warfare
    1. Allowing machines to act without human supervision
    2. Processing and interpreting large amounts of data ‘
    3. Aiding command and control of war

The main arguments against (lethal) autonomous weapon systems are that they support extrajudicial killings, take responsibility away from humans, and make wars or killings more likely

Robot and Morality

- Using robots in warfare has several potential advantages
  - Very efficient … need few human controller to monitor of the use of thousands of autonomous weapons

  - Robots are more ethical than humans

  - Why?

Robot and Morality

- Moral issues
  - Another question is whether using autonomous weapons in war would make wars worse, or make wars less bad.

AI and Morality

- Does AI have moral agency?
  - There has been a steady increase in the number of people killed by robots or other AI-systems
    - Autonomous vehicles
    - Robots in a factory
  - Why AI is perceived as morally responsible?
  - Who has moral responsibility for these and other AI-based tragedies?
    - Is it the AI system?
    - Is it the designers of the AI system? Or
    - Both?

Questions

- Can you list the strength of human intelligence and AI as well? (3 pts)

- List the main components of BCI? [2 pts]